Java Full Stack Developer (Intern)

INTERNSHIP REPORT submitted in partial fulfillment of the requirements

for the Award of the Degree of

BACHELOR OF TECHNOLOGY

In

INFORMATION TECHNOLOGY

By

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Codec Technologies

Java Full Stack Developer Intern (Codec Technologies)

(Mode: Online)

DEPARTMENT OF INFORMATION TECHNOLOGY

V R SIDDHARTHA ENGINEERING COLLEGE

(AUTONOMOUS - AFFILIATED TO JNTU-K, KAKINADA)

Approved by AICTE &Accredited by NBA

KANURU, VIJAYAWADA-520007

ACADEMIC YEAR

(2025-26)

V.R.SIDDHARTHA ENGINEERING COLLEGE

(Affiliated to JNTUK: Kakinada, Approved by AICTE, Autonomous)
(An ISO certified and NBA accredited institution)
Kanuru, Vijayawada – 520007

CERTIFICATE

This is to certify that this internship report titled **"Java Full Stack Developer Internship (Codec Technologies)"** is a bonafide record of work carried out by **AMARNATH ANKEM** (228W1A1269) during their internship at **Codec Technologies** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26.

Supervisor name	(**Dr. M. Suneetha)**
Position	Dean R&D, Professor &Head
Company	Dept. of Information Technology

Date of examination: **EXAMINER SIGNATURE**

DEPARTMENT OF INFORMATION TECHNOLOGY

V.R.SIDDHARTHA ENGINEERING COLLEGE

PROJECT SUMMARY

S.No	Item	Description
1	**Internship Title**	**Java Full Stack Developer Internship (Codec Technologies)**
2	**Student Name & Roll Number**	Amarnath Ankem (228W1A1294)
3	**Internship Supervisor/ Guide**	Dr. M Ramesh
4	**Intern Role**	Java Full Stack Developer Intern
5	**Domain/ Area**	Java Full Stack Development
6	**Application Area**	Online Quiz System
7	**Objective of the Project**	To develop a secure and efficient Online Quiz System that enables timed quiz conduction, automatic evaluation, structured result management, and admin-controlled quiz operations using HTML, CSS, JavaScript, Node.js, Express.js, and MySQL.
8	**Internship Project Outcomes**	1. Built a complete Online Quiz System using HTML, CSS, JavaScript, Node.js, Express.js, and MySQL. 2. Enabled secure and efficient timed quiz conduction, automatic evaluation, and result generation. 3. Improved the accuracy, transparency, and management of online examinations through admin-controlled quiz operations and structured database storage.

Student Signature

1.

ACKNOWLEDGEMENT

First and foremost, I sincerely salute our esteemed institution **V.R SIDDHARTHA ENGINEERING COLLEGE** for providing me with the opportunity to undergo this internship, which has greatly enhanced my practical knowledge and professional skills. I am grateful to our principal **Dr. A.V.RATNA PRASAD**, for his encouragement and support.

On the submission of this Internship Report, I would like to extend my honour to **Dr. M.Suneetha**, Dean R&D, Professor & Head of the Department, IT for her constant motivation and support during the course of my work.

I feel glad to express my deep sense of gratefulness to my internship supervisor **Dr. M Ramesh** for **his** guidance and assistance in completing this internship successfully.

I would also like to convey my sincere indebtedness to all faculty members, including supporting staff of the Department, friends and family members who bestowed their great effort and guidance at appropriate times without which it would have been very difficult on my part to finish the project work.

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ABSTRACT

This project focuses on the design and development of a secure and efficient Online Quiz System using HTML, CSS, JavaScript, Node.js, Express.js, and MySQL. The main objective of the system is to provide a digital platform for conducting quizzes in a structured, reliable, and user-friendly manner. The project emphasizes server-controlled timed quizzes, secure storage of quiz data, automatic evaluation, and result management through a backend-intensive architecture. Key components of the system include a responsive frontend interface for students and administrators, an Express.js-based backend server for handling application logic, and a MySQL relational database for storing users, quizzes, questions, answers, attempts, and results.

The system was designed to demonstrate efficient quiz management through proper integration of frontend, backend, and database technologies. The workflow includes student login, quiz selection, timed quiz participation, automatic submission, score calculation, and result generation. On the administrative side, the system supports quiz creation, question management, and performance monitoring. The use of a normalized database structure with appropriate relationships, joins, and constraints ensures data consistency, security, and efficient retrieval of information.

This project strengthened practical knowledge in full-stack web development, backend logic design, database management, authentication handling, and result processing. The final outcome is a fully functional and industry-relevant quiz platform capable of supporting real-world academic and institutional assessment needs. Overall, the work provides a strong foundation in backend-intensive application development and demonstrates the ability to build scalable, database-driven web solutions. In addition, the project highlights the importance of digital examination systems in improving efficiency, accuracy, and transparency in the assessment process. By automating quiz conduction, evaluation, and record maintenance, the system reduces manual effort and enhances reliability. This project serves as an example of how modern web.

Key-words: Online Quiz System, Node.js, Express.js, MySQL, HTML, CSS, JavaScript, Timed Quiz, Automatic Evaluation, Result Management, Database Design, Web Application, Admin Dashboard.

CHAPTER-1

Introduction

This internship, conducted under Codec Technologies, provided a valuable opportunity to gain practical experience in full-stack web development through the design and implementation of a real-world software application. As part of the internship, I worked on developing an Online Quiz System, a backend-intensive web application that supports quiz creation, timed quiz conduction, automatic evaluation, and result management. The project was developed using HTML, CSS, JavaScript, Node.js, Express.js, and MySQL, enabling me to understand how frontend interfaces, backend logic, and relational databases work together in a complete application environment.

Through this project, I gained hands-on exposure to important concepts such as client-server architecture, database connectivity, authentication, server-side validation, and structured data handling. The internship helped bridge the gap between theoretical knowledge and practical implementation by allowing me to build a complete web-based system from scratch. Overall, this experience improved my technical skills, strengthened my confidence in full-stack development, and provided a strong foundation for future academic and professional work in web application development.

1.1 Company Overview

Codec Technologies is a training and technology-focused organization that provides internship and skill development opportunities in domains such as **full-stack development, web technologies, software engineering, and application development**. The company focuses on helping students and learners gain practical exposure to industry-relevant tools, frameworks, and development practices through project-oriented training programs. By combining conceptual learning with real-time implementation, Codec Technologies supports learners in building technical confidence and improving their readiness for professional roles in the IT industry.

The organization emphasizes hands-on learning by encouraging interns to work on complete projects that involve real-world problem solving, structured development workflows, and modern software tools. Through its internship programs, Codec Technologies helps students strengthen their knowledge in both frontend and backend technologies while also improving their ability to design, develop, and deploy functional applications.

1.2 Internship Role and Duration

I participated in the Full Stack Development Internship offered by Codec Technologies, where my role involved designing and developing a complete Online Quiz System. During the internship, I was responsible for building the frontend interface, developing the backend server, designing the relational database, and integrating all modules into a fully functional web application. My work included implementing timed quizzes, secure data handling, automatic score calculation, result storage, and admin-level quiz management features.

The internship was carried out over the assigned training duration, during which I completed the project in a structured and practical manner. This project-based internship approach helped me gain direct exposure to full-stack development practices, backend-intensive application design, and database-driven system implementation. It also improved my understanding of how to build real-world web applications that are secure, scalable, and user-friendly.

1.3 Objectives of the Internship

1. To apply theoretical knowledge of full-stack web development by building a real-world Online Quiz System using HTML, CSS, JavaScript, Node.js, Express.js, and MySQL.
2. To develop a secure and efficient system that supports timed quiz conduction, automatic evaluation, and structured result management.
3. To gain hands-on experience in backend development, database design, authentication, server-side validation, and frontend-backend integration.
4. To improve practical skills in designing database-driven web applications with proper user management, admin controls, and reliable data handling mechanisms.

CHAPTER 2

TOOLS AND TECHNOLOGIES USED

The internship involved the use of several web development tools, programming technologies, and database systems to design and implement a complete Online Quiz System. These technologies supported the development of a secure, responsive, and backend-intensive application capable of handling timed quizzes, user interactions, automatic evaluation, and result management. The frontend was developed using HTML, CSS, and JavaScript to create an interactive user interface, while Node.js with Express.js was used for backend logic and server-side operations. MySQL was used as the relational database for storing users, quizzes, questions, answers, and results. In addition, development tools such as Visual Studio Code, MySQL Workbench, and Postman were used for coding, database management, and API testing. Together, these tools enabled the successful development and integration of a full-stack web application that reflects real-world industry practices.

2.1 Software / Platforms

1. **HTML, CSS, and JavaScript:**

   HTML, CSS, and JavaScript were used to build the frontend interface of the Online Quiz System. HTML was used to structure the web pages, CSS was used for styling and layout design, and JavaScript was used to add interactivity and dynamic behavior. These technologies helped create user-friendly interfaces for both students and administrators.

2. **Node.js:**

   Node.js was used as the runtime environment for executing JavaScript on the server side. It enabled the development of backend functionality such as request handling, quiz processing, authentication support, timer control, and communication between the frontend and database.

3. **Express.js:**

   Express.js was used as the backend web framework for building the server-side application. It simplified routing, request-response handling, middleware integration, and API development. Express.js played an important role in implementing quiz logic, user operations, admin controls, and result processing.

4. **MySQL:**

   MySQL was used as the relational database management system for storing and managing the application data. It maintained structured information such as student details, admin data, quizzes, questions, options, quiz attempts, and results. MySQL also supported joins, constraints, and normalized table design for efficient data storage and retrieval.

5. **MySQL Workbench:**

MySQL Workbench was used for database creation, table design, query execution, and data management. It helped in developing the database schema, inserting sample data, and verifying stored quiz and result records.

6. **Postman:**

Postman was used for testing backend routes and verifying API functionality during development.

2.2 Programming Languages / Frameworks

• HTML:

HTML was used to create the basic structure of the web pages in the Online Quiz System. It helped in designing the layout for student pages, admin pages, quiz forms, login sections, and result displays.

• CSS:

CSS was used to style the application and improve the visual appearance of the system. It helped in creating a clean, responsive, and professional user interface for both students and administrators.

• JavaScript:

JavaScript was used to add interactivity and dynamic behavior to the frontend of the application. It supported functions such as form handling, timer display, validation, and smooth user interaction during quiz participation.

• Node.js:

Node.js was used as the server-side runtime environment for executing backend logic. It handled requests from the frontend, managed quiz operations, processed submissions, and connected the application with the MySQL database.

• Express.js:

Express.js was used as the backend framework to simplify routing, middleware handling, and API development. It played a major role in implementing authentication, timed quiz handling, result generation, and admin-level operations.

• MySQL:

SQL was used in MySQL for creating tables, inserting records, updating quiz data, and retrieving results. It also supported joins, constraints, and structured queries required for efficient database management.

CHAPTER 3:

WORK DESCRIPTION

During this project, my primary focus was on developing a complete Online Quiz System as a backend-intensive web application. I was involved in designing and implementing the full workflow of the system, including frontend interface development, backend logic creation, database design, timed quiz handling, result processing, and admin-level management. The project was built using HTML, CSS, JavaScript, Node.js, Express.js, and MySQL, which allowed me to understand how different layers of a full-stack application interact in a real-world environment.

The tasks included creating user-friendly interfaces for students and administrators, developing server-side functionality for quiz control and validation, storing application data in a relational database, and implementing automatic score calculation and result generation. This project provided hands-on experience in full-stack development, relational database handling, authentication concepts, and backend-driven application workflows. It also strengthened my practical skills in building secure, structured, and scalable web applications.

3.1 Overview of the Project / Task Assigned

During this project, I was assigned the task of designing and developing a complete Online Quiz System to strengthen my practical skills in full-stack web development and backend-intensive application design. The main objective was to build a secure, efficient, and user-friendly platform for conducting quizzes and managing results in a structured manner. The tasks included:

- Developing the frontend interface for students and administrators using HTML, CSS, and JavaScript.
- Building the backend application using Node.js and Express.js to handle quiz logic, validation, request processing, and result generation.
- Designing and implementing a MySQL relational database to store users, quizzes, questions, options, attempts, and results.
- Creating a server-controlled timed quiz system to ensure fairness and proper quiz conduction.
- Implementing automatic score calculation and storing quiz results for future reference.
- Developing an admin management module for quiz creation, question management, and monitoring student performance.

Each task was designed to improve understanding of application flow, database-driven system design, and backend processing. This project provided hands-on exposure to real-world web development practices and strengthened my practical knowledge of frontend-backend integration, data handling, and system management.

3.2 System Workflow / Architecture

1.Requirement Analysis:

The project began with understanding the main requirements of the system, such as secure quiz conduction, timed assessment, admin control, automatic evaluation, and result storage. Based on these requirements, the overall architecture of the Online Quiz System was planned.

2. Frontend Interface Development:

The user interface was developed using HTML, CSS, and JavaScript. Separate pages were designed for students and administrators to ensure smooth interaction with the system. The frontend manages quiz display, form submission, timer visibility, and result presentation.

3.Backend Processing:

The backend was developed using Node.js and Express.js, which handle application logic, request processing, response generation, authentication support, and communication between the frontend and database. It acts as the main control layer of the system.

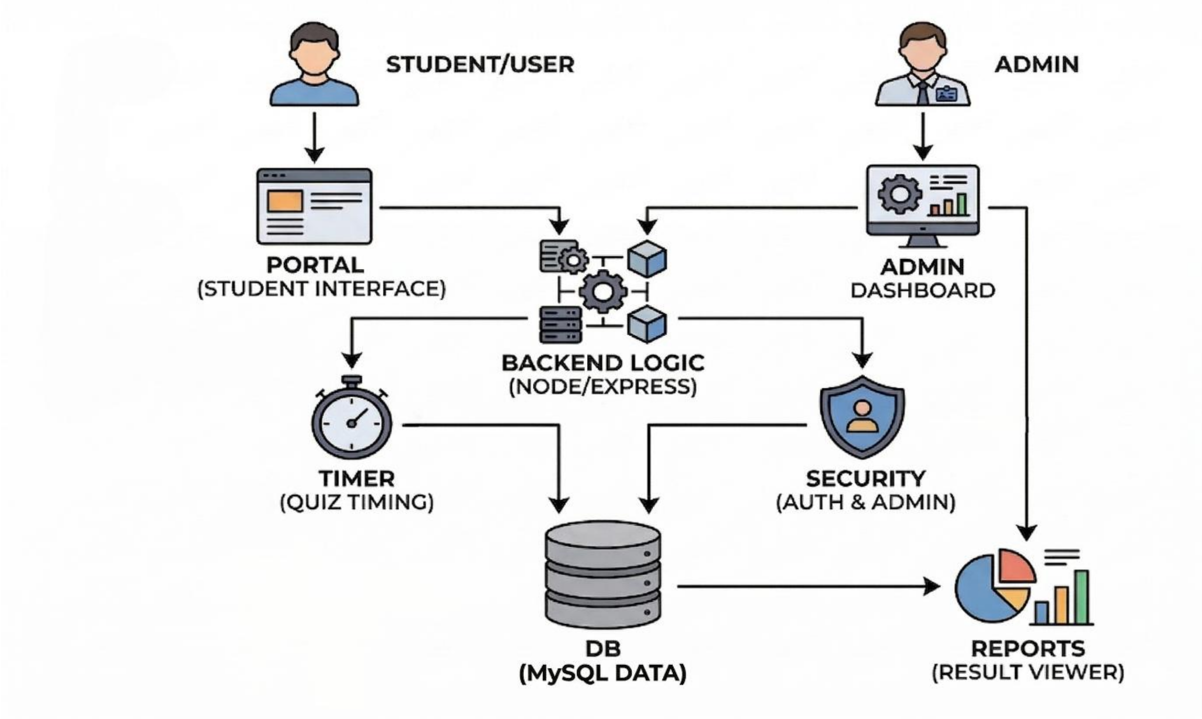

Figure 3.1 Workflow of Project

4. Database Management:

A MySQL relational database was used to store and manage all application data, including user details, quizzes, questions, answer options, attempts, and results. Proper table relationships and constraints were maintained for structured data handling.

5. Timed Quiz Execution:

When a student starts a quiz, the backend loads the quiz questions and manages the server-controlled timer. This ensures fairness in quiz conduction and prevents misuse from the client side.

6. Automatic Evaluation and Result Generation:

After submission or timeout, the backend compares student responses with the correct answers stored in the database. The system then calculates the score automatically, stores the result, and displays it to the student.

7. Admin Monitoring and Control:

The admin module allows administrators to create quizzes, add questions, manage quiz details, and monitor student performance. This helps in centralized and efficient control of the complete quiz system.

3.3 Features / Modules Implemented

1. Student Quiz Module:

- Allows students to access and attend available quizzes.
- Displays quiz questions, options, and timer in an organized format.
- Supports quiz submission and automatic result generation.

2. Timed Quiz Management Module:

- Implements a server-controlled timer for each quiz.
- Ensures fair quiz conduction with proper time restrictions.
- Supports automatic submission when the time limit is reached.

3. Admin Management Module:

- Allows admin login and secure access to management features.
- Supports quiz creation, updating, publishing, and deletion.
- Enables question and answer management within each quiz.

4. Database Management Module:

- Stores users, quizzes, questions, answer options, attempts, and results.
- Maintains data consistency using proper table relationships and constraints.
- Supports efficient retrieval and storage of quiz-related information.

5. Result Processing Module:

- Automatically evaluates submitted answers.
- Calculates the final score based on correct responses.
- Stores quiz results for future reference and performance tracking.

6. User Interface Module:

- Provides a clean and interactive interface for both students and administrators.
- Improves usability through structured pages, forms, buttons, and result views.
- Ensures a smooth overall experience while using the system.

3.4 Your Contribution / Responsibilities

- Designed and developed the complete Online Quiz System from frontend to backend and database integration.
- Created the student and admin interfaces using HTML, CSS, and JavaScript for smooth user interaction.
- Built the backend server using Node.js and Express.js to handle quiz flow, request processing, validation, and result generation.
- Designed and managed the MySQL database schema for storing users, quizzes, questions, answer options, attempts, and results.
- Implemented the server-controlled timed quiz mechanism to ensure fair and structured quiz conduction.
- Developed the automatic score calculation and result persistence module for efficient evaluation.
- Built the admin management module for quiz creation, question management, and result monitoring.
- Tested and integrated all modules to ensure smooth workflow, accuracy, and reliable system performance.

CHAPTER 4

RESULTS AND OBSERVATIONS

The project provided an opportunity to develop a complete Online Quiz System, helping bridge the gap between theoretical knowledge and practical full-stack implementation. The results demonstrate the successful integration of frontend, backend, and database technologies to conduct quizzes, manage data, and generate results efficiently. The system ensures secure quiz handling, automatic evaluation, and structured result storage, showcasing a real-world backend-intensive web application workflow.

4.1 Output / Screenshots / Demonstration

The project produced multiple outputs across different stages of the system, demonstrating the successful implementation of quiz functionality, backend processing, and database operations. The following outputs represent the working system:

- **Student Quiz Interface:**

  The quiz interface displays questions, options, and a timer for users. It allows students to interact with the system and attempt quizzes in a structured manner.

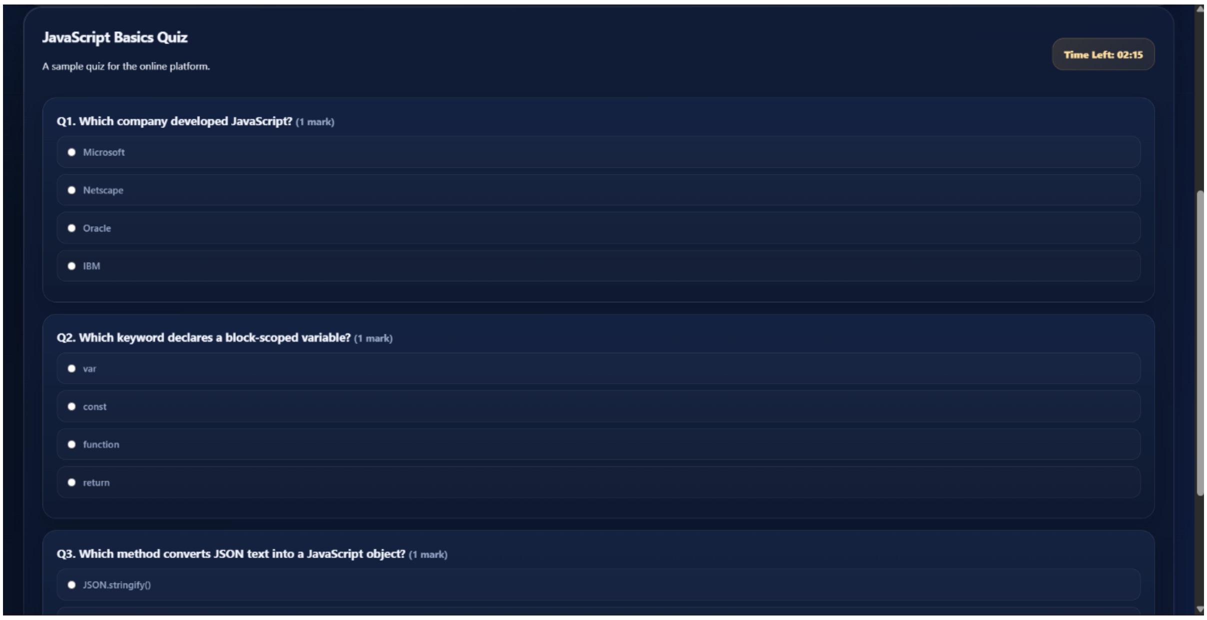

Figure 4.1: Student Quiz Interface with Timer

- **Quiz Execution with Timer:**
  The system successfully implements a server-controlled timer, ensuring that quizzes are completed within the specified time limit. This confirms proper backend control and fairness in quiz conduction.

- **Result Generation Output:**
  Raw After submission, the system automatically evaluates answers and displays the

final score. This verifies the correct implementation of the score calculation and result processing module.

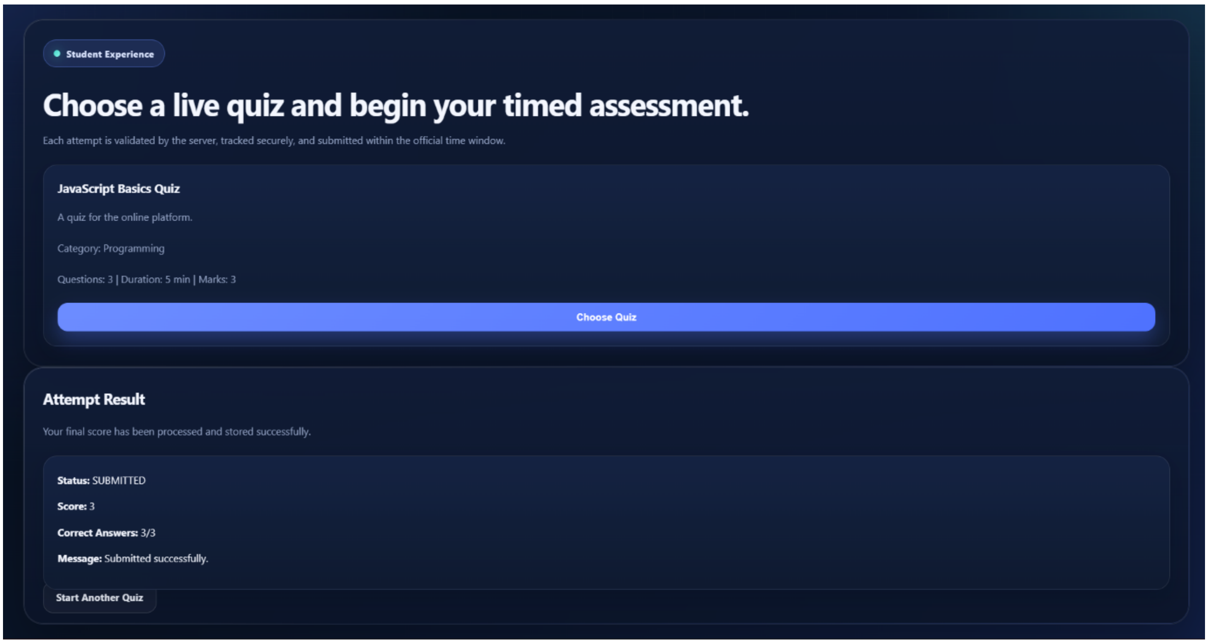

Figure 4.2: Quiz Result / Score Output

- **Admin Dashboard:**
  The admin panel allows administrators to manage quizzes, view results, and control system operations. This demonstrates successful implementation of the admin management module.

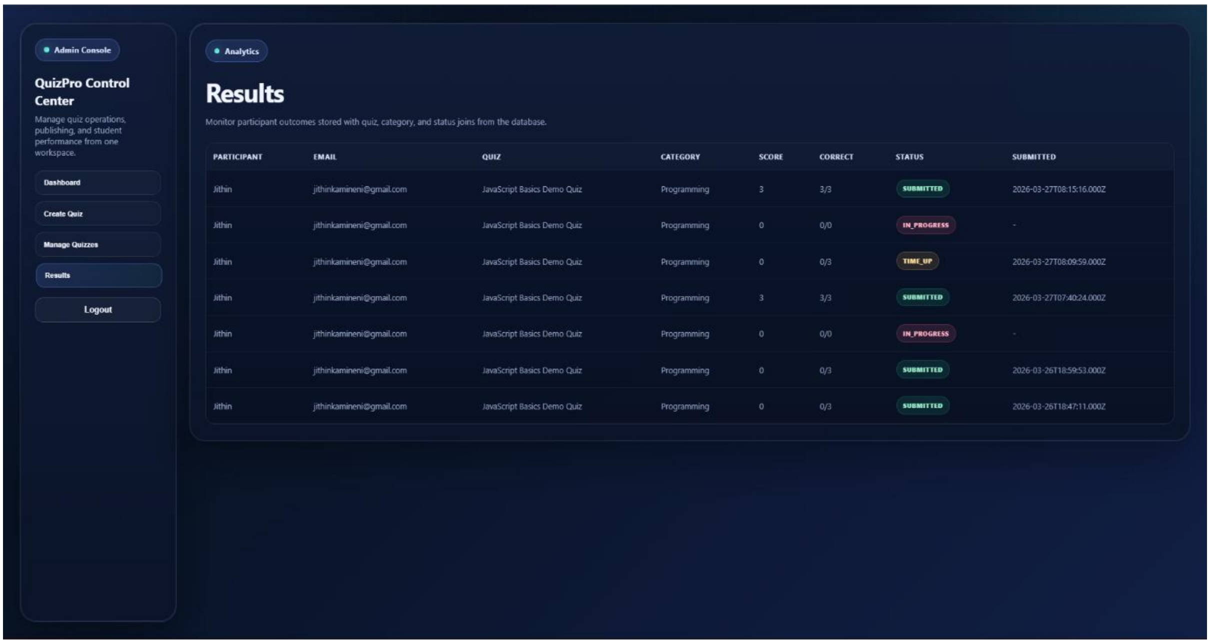

Figure 4.3: Admin Dashboard Interface

- **Quiz / Question Management:**
  The system allows admins to create and manage quizzes and questions, confirming proper backend integration and database interaction.

4.2 Skills Acquired

- Developed strong skills in full-stack web development using HTML, CSS, JavaScript, Node.js, and MySQL.
- Gained practical experience in backend development and database integration.
- Learned to implement server-controlled logic such as timed quizzes and validation.
- Improved knowledge of database design, SQL queries, and relational data management.
- Enhanced ability to build user-friendly interfaces and admin dashboards.
- Strengthened problem-solving and debugging skills during system development.

4.3 Challenges Faced and Solutions

During the development of these projects, several challenges were encountered and resolved:

• **Challenge 1:** Managing timed quiz execution

Solution: Implemented a server-controlled timer to ensure fairness and prevent manipulation from the client side**.**

• **Challenge 2:** Handling data storage and relationships

Solution: Designed a normalized MySQL database schema with proper relationships and constraints.

• **Challenge 3:** Automatic result calculation

Solution: Developed backend logic to compare answers and generate scores instantly after submission.

• **Challenge 4:** Admin control and data management

Solution: Built an admin dashboard to manage quizzes, questions, and results efficiently.

• **Challenge 5:** Improving user interface

Solution: Enhanced UI design to make the system more professional and user-friendly.

CHAPTER 5

CONCLUSION AND FUTURE STUDY

The project provided hands-on experience in developing a complete Online Quiz System using HTML, CSS, JavaScript, Node.js, Express.js, and MySQL. I worked on building a backend-intensive web application that supports quiz creation, timed quiz conduction, automatic evaluation, result management, and admin-level control. Through this project, I enhanced my skills in frontend development, backend logic implementation, database design, authentication handling, and full-stack integration. The experience improved my understanding of real-world web application workflows, structured data management, and server-side processing. Future improvements could include adding user authentication with advanced security, improving analytics and reporting features, supporting multiple quiz categories, and making the platform more scalable and responsive. Overall, this project strengthened my technical capabilities and prepared me for more advanced academic and professional opportunities in full-stack web development.

5.1 Summary of Internship Experience

This internship offered practical exposure to full-stack web development through the implementation of a real-world Online Quiz System. Throughout the project, I worked on important stages of application development, including frontend design, backend processing, database integration, timed quiz management, result generation, and admin-level operations.

The project began with understanding the system requirements and planning the structure of the application. I then developed the frontend interface using HTML, CSS, and JavaScript, which helped me understand how to create interactive and user-friendly pages for both students and administrators. After that, I worked on the backend using Node.js and Express.js, where I implemented routing, validation, quiz logic, result processing, and communication with the database.

A major part of the project involved designing a MySQL relational database to store quiz details, questions, answer options, user data, attempts, and results. This improved my understanding of normalized database design, SQL queries, joins, and constraints. I also gained practical experience in implementing a server-controlled timer, automatic score calculation, and structured result storage. Overall, the internship strengthened my skills in web development, database-driven system design, debugging, and project integration, and it prepared me for future roles in software and application development.

5.2 Future Scope / Improvements

The Online Quiz System can be further enhanced in several ways:

- **Advanced Authentication and Security:**
  Future versions can include secure login systems using JWT, password encryption, session management, and role-based access control.

- **Performance Analytics:**
  The system can be extended to provide detailed student performance analysis, charts, and progress reports for better academic monitoring.
- **Multiple Quiz Categories and Difficulty Levels:**
  Additional features can be added to support different subjects, quiz categories, and varying difficulty levels for a richer learning experience.
- **Responsive and Mobile-Friendly Design:**
  The interface can be further improved to ensure seamless use across mobile phones, tablets, and desktop devices.
- **Real-Time Notifications and Auto-Save:**
  Future improvements can include quiz reminders, submission alerts, and automatic saving of answers during quiz attempts.
- **Scalability and Deployment:**
  The system can be deployed on cloud platforms and enhanced to support larger numbers of users and simultaneous quiz attempts.